S3 | R53 | CDN

1. Configure VPC peering in cross regions.

For testing purpose,

I have created an EC2 instance using default VPC of Virginia region.

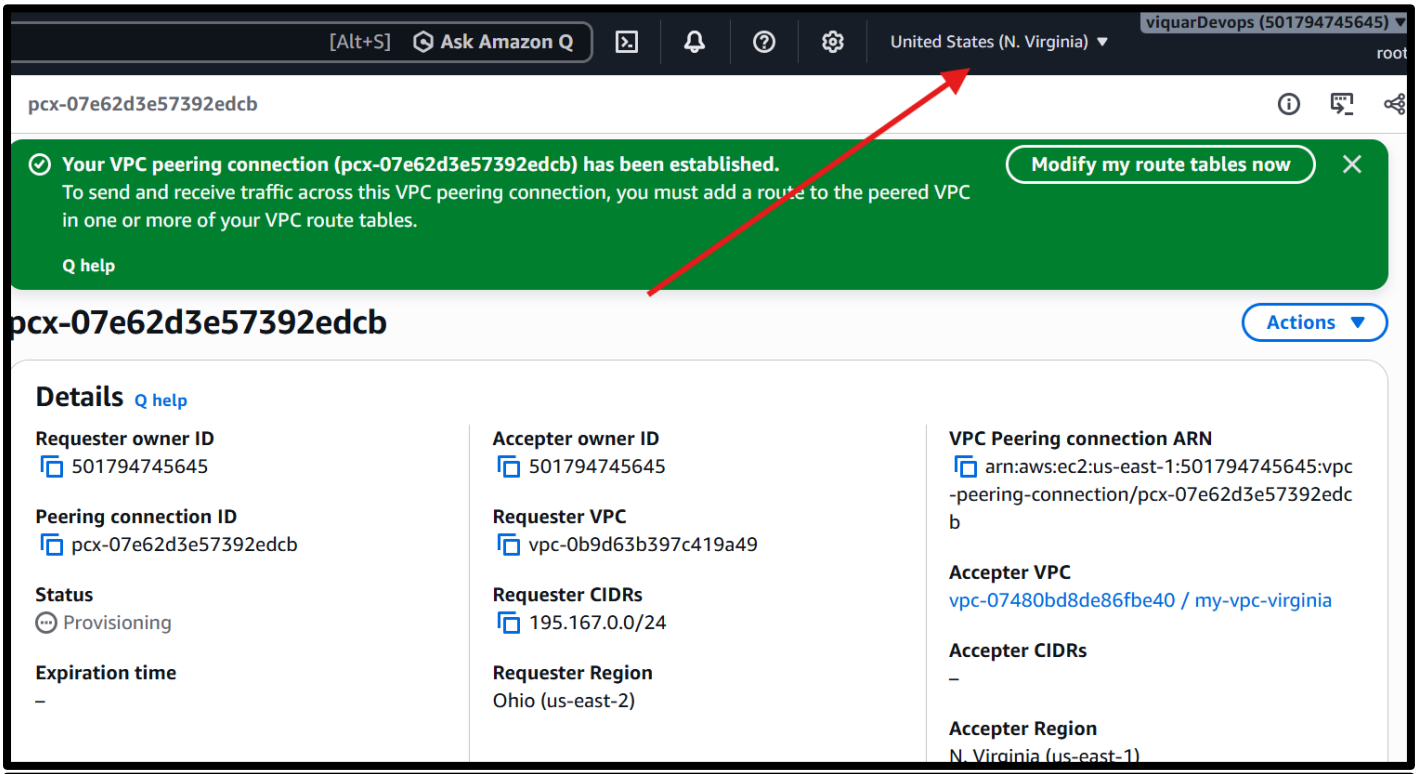
I have one EC2 already running in Ohio region with my new VPC.

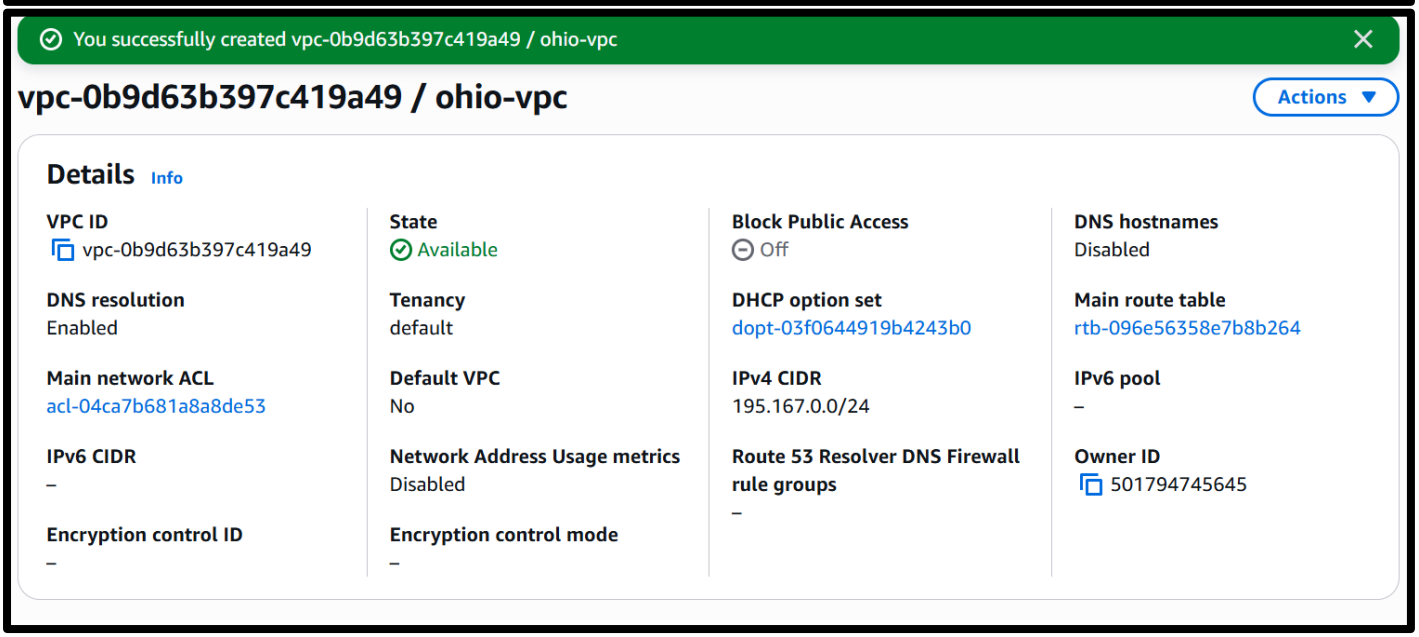
I tried to ping Ohio regions EC2 using public ip so its success but using private IP it not connection that's why I will do VPC peering.

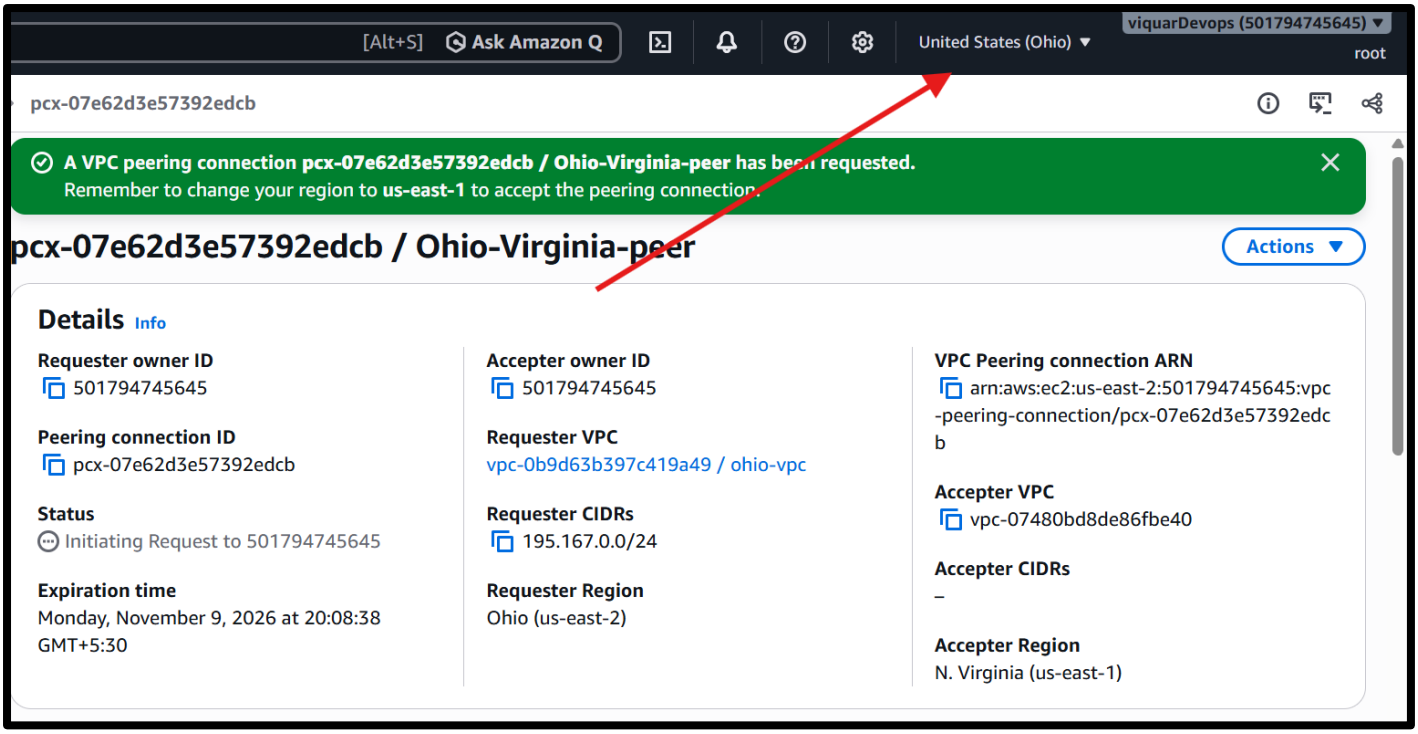
I have created a Peering request for peering with VPC in Virginia from Ohio below are the steps.

VPC peering →create a peering connection →add name → source VPC select → region of destination → and VPC ID of destination → accept if it ends user is me only.

Configure route table in which in Ohio add the CIDR rang of VPC of Virginia and in Virginia add CIDR range of Ohio select peering connection







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2. Purchase one domain from GoDaddy.

Steps to Perform:

Open a web browser (Google Chrome). → Visit the official website: https://www.godaddy.com → Create an account or log in to your existing GoDaddy account. → Enter your desired domain name in the search bar. → Click on Search Domain. → Check whether the domain name is available. →Select the available domain extension (such as .com, .in, or .org). → Click on Add to Cart. → Proceed to checkout. → Select a payment method. → Complete the payment process. → Verify the purchase confirmation from GoDaddy.

→ Open My Products in your GoDaddy account. → Verify that the purchased domain appears under your products.

Expected Outcome:

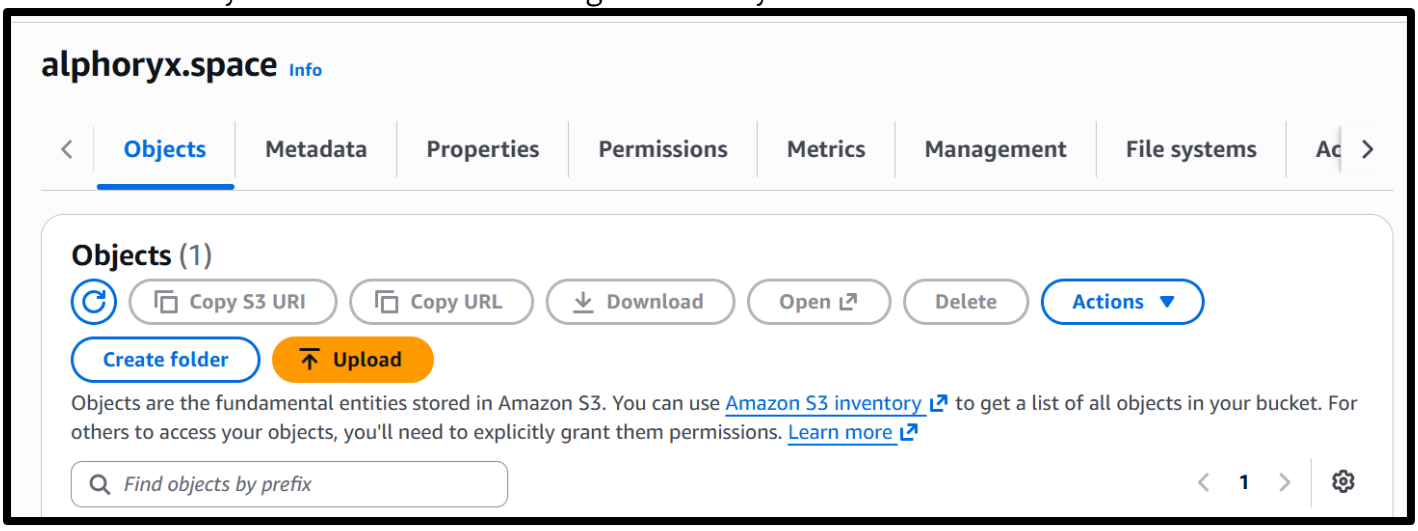
- Successfully purchase a domain name.
- Domain gets registered under your GoDaddy account.
- Domain becomes available for DNS configuration and future website hosting.

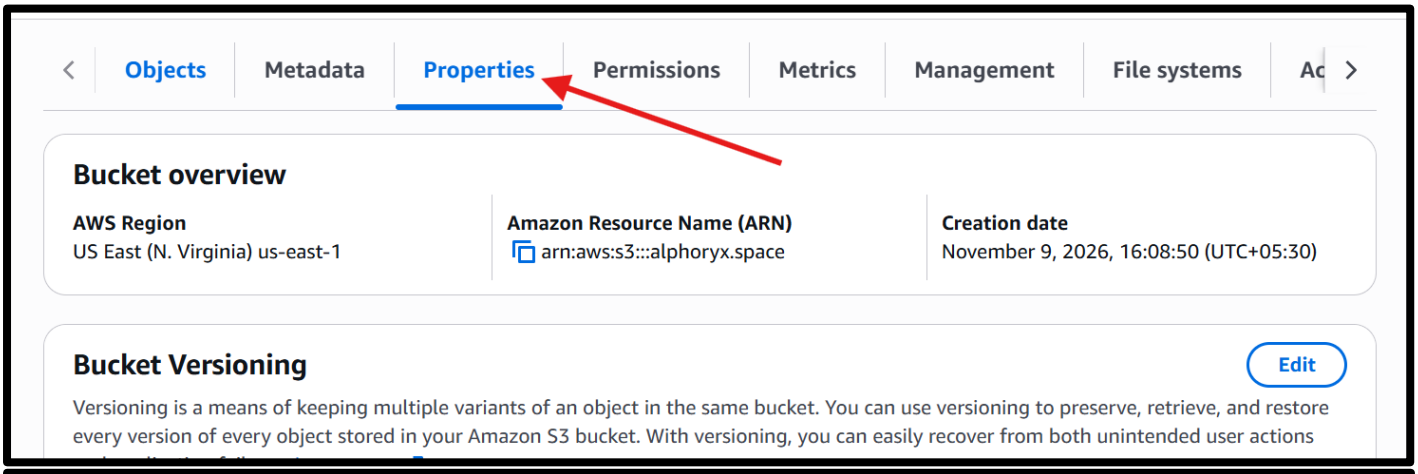
Result: Successfully purchased and registered a domain name using GoDaddy.

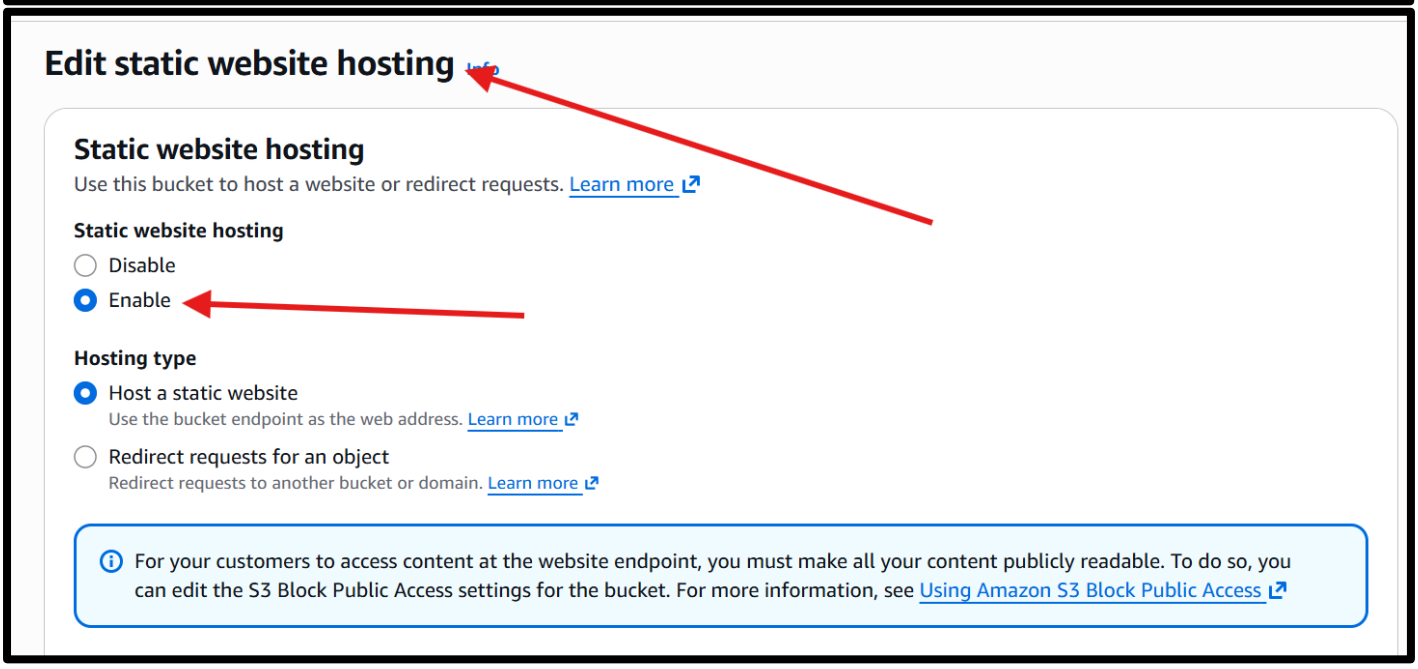
3. Deploy static website in S3.

Steps to Perform:

→ Log in to AWS Console and search for S3. → Create a bucket with a unique name and select the AWS Region. → Configure public access settings and create the bucket. → Open the bucket and upload website files (index.html, CSS, JS, images). Go to Properties → Static website hosting → Edit.
→ Enable static website hosting and select index.html as the index document. → Save the changes and configure the bucket policy for public read access. → Copy the S3 website endpoint and open it in a browser → Verify that the website is loading successfully.







Specify the home or default page of the website.

index.html

Error document - *optional*

This is returned when an error occurs.

error.html

Redirection rules – *optional*

Redirection rules, written in JSON, automatically redirect webpage requests for specific content. Learn more

Successfully edited static website hosting.

Use this bucket to host a website or redirect requests. Learn more

We recommend using AWS Amplify Hosting for static website hosting

Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about Amplify Hosting or View your existing Amplify apps

Create Amplify app

S3 static website hosting

Enabled

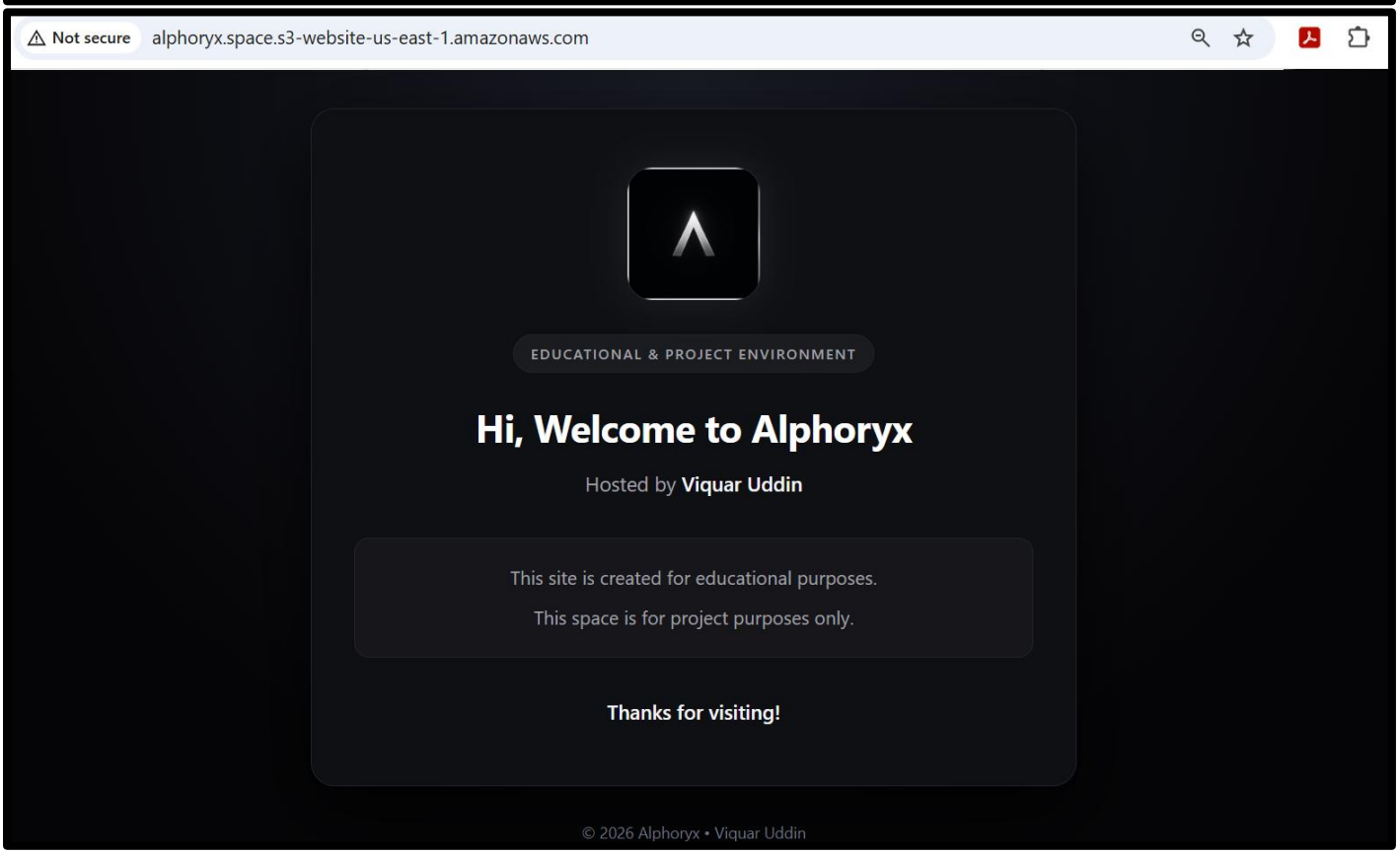
Hosting type

Bucket hosting

Bucket website endpoint

When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. Learn more

http://alphoryx.space.s3-website-us-east-1.amazonaws.com

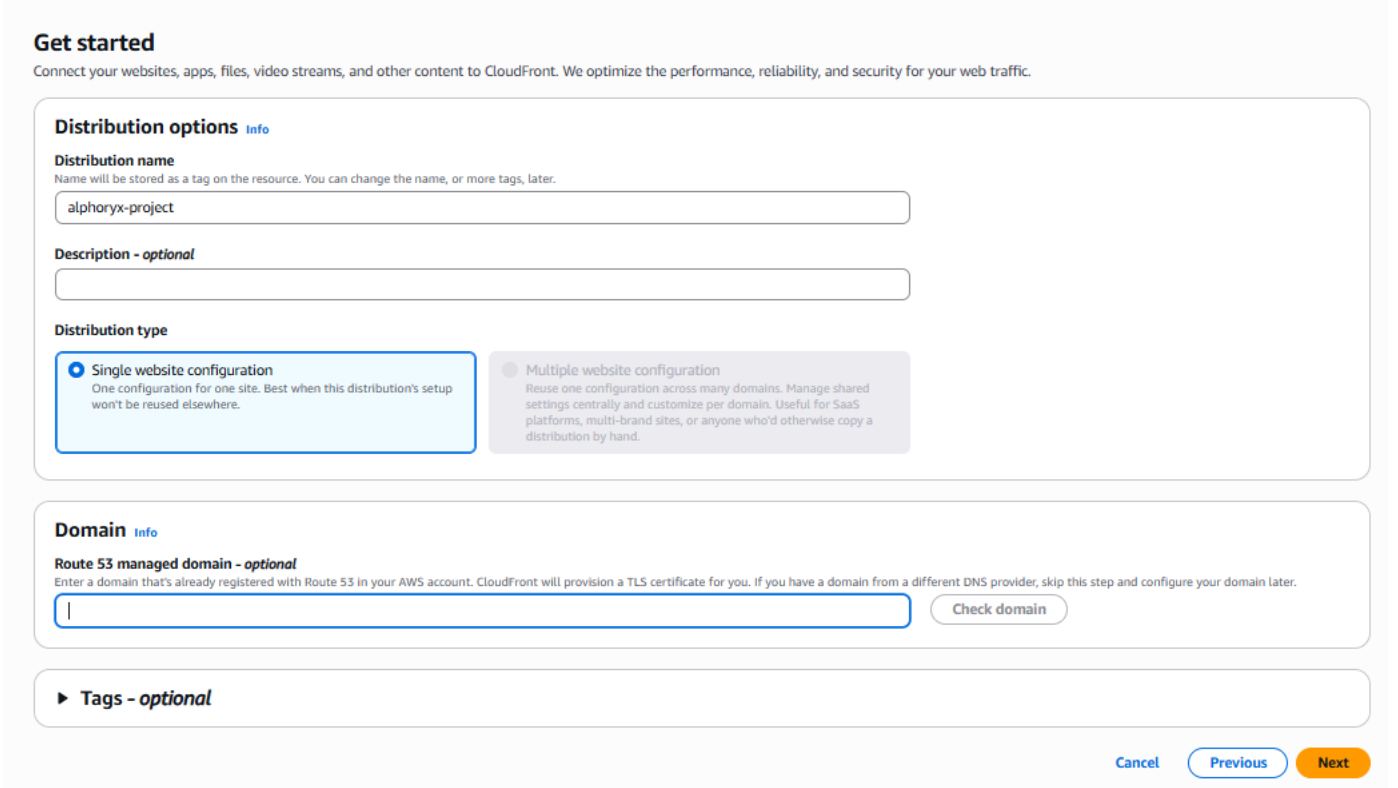


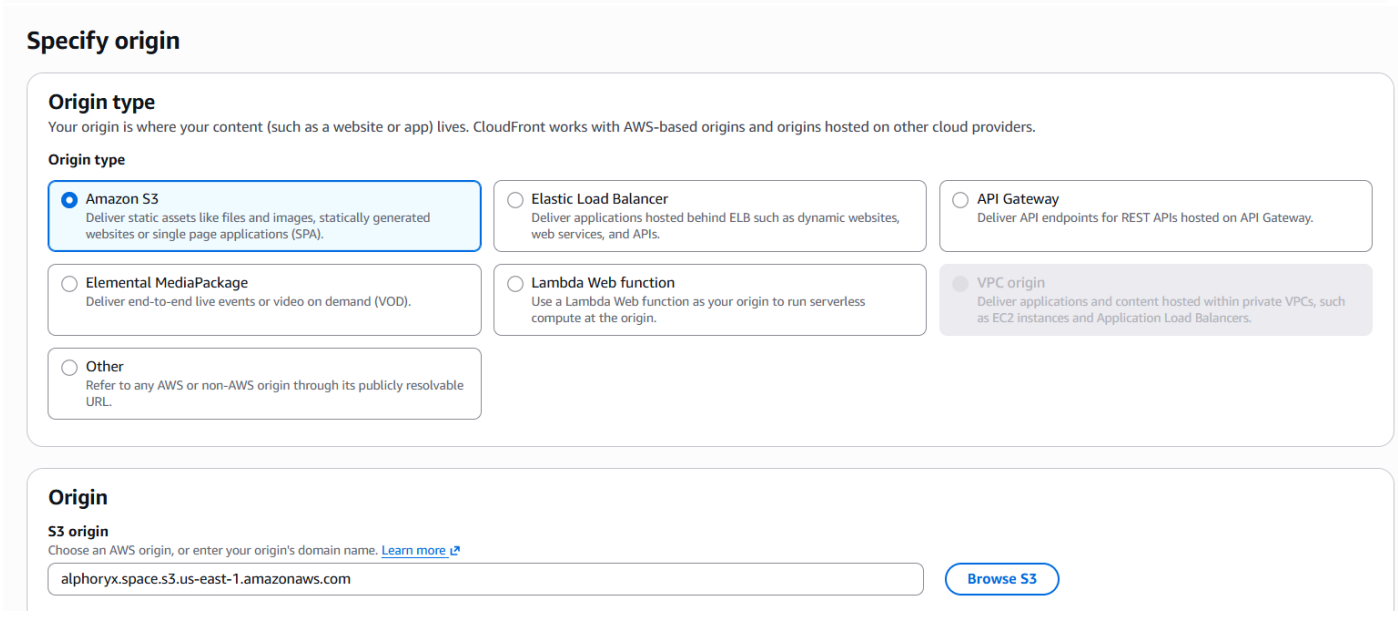
4. Create a CDN and attach one SSL certificate.

Request an SSL Certificate → Open AWS Management Console → Search for AWS Certificate Manager (ACM) → Select region US East (N. Virginia – us-east-1) → Click Request a Certificate → Choose Request a Public Certificate → Click Request.

Validate the SSL Certificate → Open the requested certificate in ACM → Copy the DNS validation CNAME record → Open GoDaddy DNS Management → Add the copied CNAME record → Save the DNS record → Wait until certificate status changes to Issued

Create a CloudFront Distribution → Open AWS Management Console → Search for Amazon CloudFront → Click Create Distribution → Select your origin (S3 Bucket or Web Server) → Configure default cache settings → Set Viewer Protocol Policy to Redirect HTTP to HTTPS





Settings Info

CloudFront provides default origin and cache settings based on what origin you selected. View default settings for S3

Allow private S3 bucket access to CloudFront Info

CloudFront will update your S3 bucket policy to allow CloudFront to access your S3 bucket. The policy allows CloudFront to access the bucket only when the request is on behalf of the CloudFront distribution that contains the S3 origin.

☑ Allow private S3 bucket access to CloudFront - *Recommended*

Origin settings

Origin settings control how CloudFront connects to the specified origin.

◉ Use recommended origin settings ○ Customize origin settings

Cache settings

Cache settings determine when CloudFront serves cached content and when it fetches new content from the origin.

◉ Use recommended cache settings tailored to serving S3 content ○ Customize cache settings

Cancel Previous Next

Enable security

Web Application Firewall (WAF) Info

Security protections from WAF are included in your plan at no additional charge.

▼ **Included security protections**

- Protect against the most common vulnerabilities found in web applications.
- Protect against malicious actors discovering application vulnerabilities.
- Block IP addresses from potential threats based on Amazon internal threat intelligence.

☐ Use monitor mode

Count how many of your requests would be blocked by this WAF configuration. When ready, you can disable monitor mode to begin blocking requests.

Protection against Layer 7 DDoS attacks Available with the Business plan Change plan

Stops DDoS attacks within seconds. AWS learns your unique application patterns within minutes of activation, accurately distinguishing between attacks and natural traffic surges.

Cancel Previous Next

Review and create

General configuration Edit

Distribution name	Description	Billing	Today's pro-rated charge (estimate)
alphoryx-project	-	Free ($0/month)	$0

Origin Edit

ⓘ Because you granted CloudFront access to your origin, CloudFront can write and update S3 bucket policies that restrict access to your S3 origin to CloudFront.

S3 origin	Origin path	Grant CloudFront access to origin	Enable Origin Shield
alphoryx.space.s3.us-east-1.amazonaws.com	-	Yes	No

Connection attempts	Connection timeout
3	10

Cache settings Edit

CloudFront will apply default cache settings tailored to serving content from a S3 origin. You can customize settings after you create your distribution.

Security Edit

Security protections	Use monitor mode
Enabled	No

Usage and transfer limits will be prorated for your first month. Subject to usage and transfer limits.

Cancel Previous Create distribution

Update on ACM public certificate validity
We have updated the validity period for ACM public certificates to comply with new CA/Browser Forum guidelines. We have also reduced exportable certificate pricing. No action is required. Learn more

Certificates (1)

More actions | Manage expiry events | Import | Request

Search certificates

1

Certificate ID	Status	Common name

Domains (1)

Create records in Route 53 | Export to CSV

1

CNAME name	CNAME value
_c7f1b7f3fde3f84286fa99ecb6028d10.alphoryx.space.	_3265400f729ff986da82f062089cb298.wzccmgtwzk.acm-validations.aws.

All Products and Services

Domains

Manage All

alphoryx.space

Protection Plan: None
Upgrade Protection

DNS | Set Up | Manage

Create a website | Set up an email account | Connect to an existing site

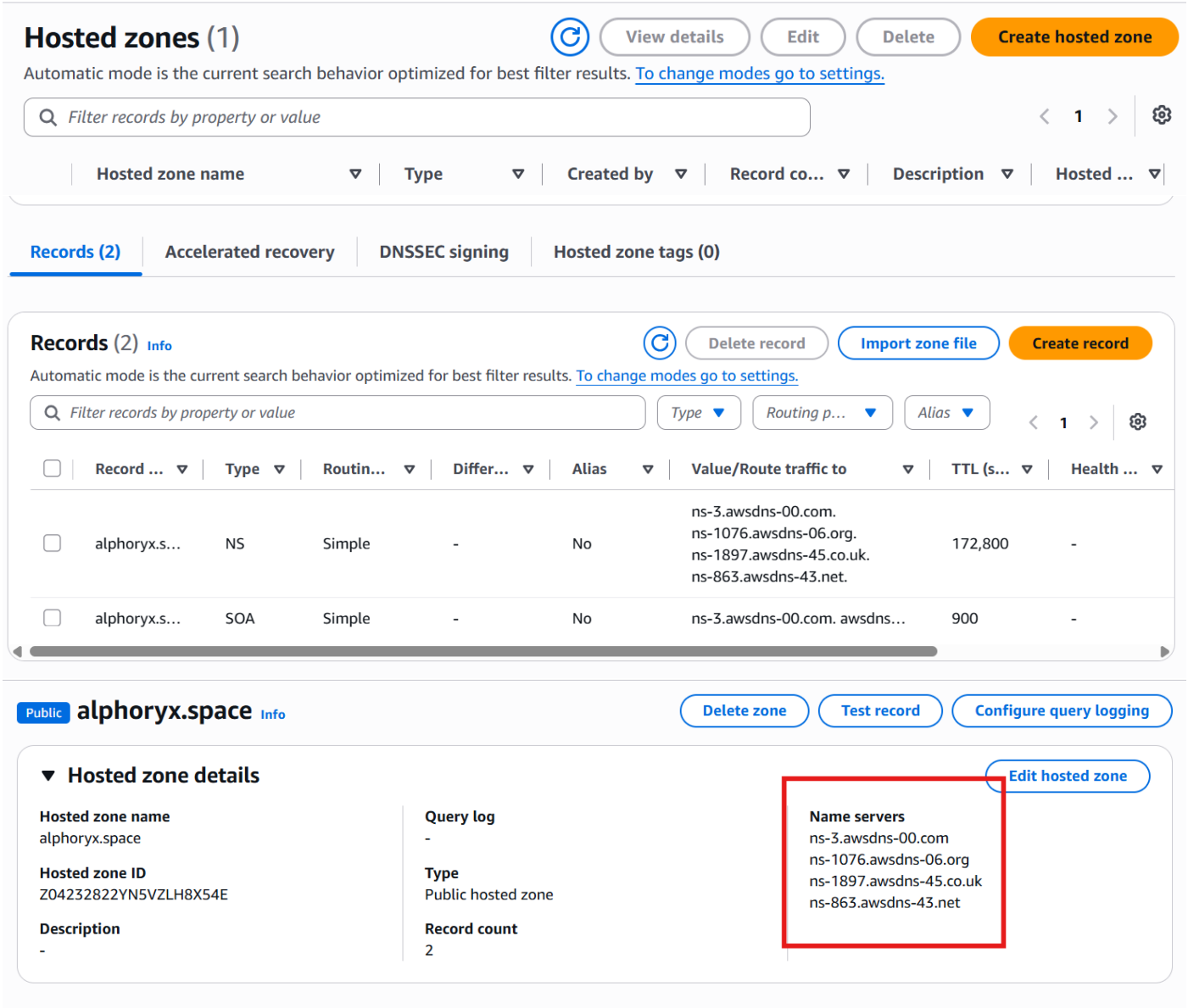
5. Create a Route 53 hosted zone and map the domain with the CDN.

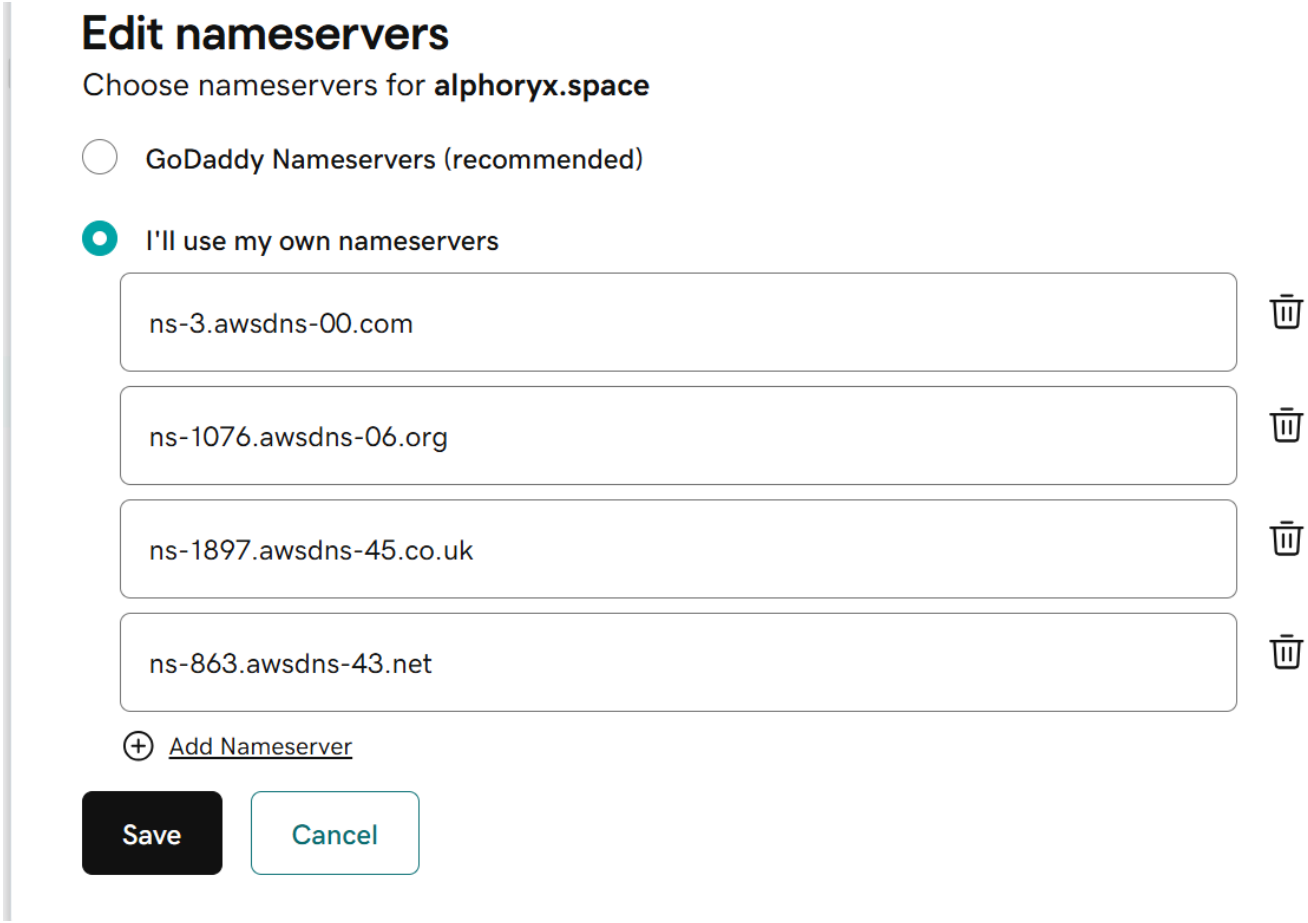
Steps:

Open the AWS Management Console and search for Route 53. → Select Hosted Zones and click Create Hosted Zone. → Enter your domain name (e.g., alphoryx.space). → Select Public Hosted Zone and create the hosted zone. → Open the created hosted zone and copy the NS (Name Server) records.

Log in to your GoDaddy account.
→ Open My Products and select your domain. → Go to DNS Management and update the existing nameservers. → Replace the GoDaddy nameservers with the four AWS Route 53 nameservers. → Save the changes and wait for DNS propagation.

Return to AWS Route 53 and verify that the domain is mapped successfully. → Once DNS propagation is complete, the domain will be managed through AWS Route 53





6. Update the index.html in the S3 bucket and ensure the updated file is accessible using the domain name.

Steps:
→ Open the AWS Management Console and search for S3. → Select the S3 bucket used for website hosting. → Open the bucket and locate the index.html file. → Upload the updated index.html file and replace the existing file.

Note : Make sure the bucket permissions and website hosting settings are configured correctly.

7. Share the domain name in Slack to test the connectivity.

https://alphoryx.space/